

1. Introduction

This implementation utilizes a ResNet-RS 200 architecture to execute a 101-class image classification task. The data pipeline incorporates a three-stage augmentation process consisting of geometric, photometric, and noise-based transformations. Geometric operations, specifically 180-degree rotations and horizontal flips, are applied to establish spatial invariance. These are followed by color jittering and Gaussian blur to account for variations in environmental lighting and image resolution. These techniques regularize the training process, facilitating the extraction of structural features that remain invariant across different orientations, lighting conditions, and focus levels.

2. Methods

a. Data Augmentation

i. Geometric and Spatial Transforms

By applying a Random Rotation with range of $\pm 180^\circ$ and a Random Horizontal Flip with a 50% probability, the pipeline enforces spatial invariance. This ensures the model recognizes a class regardless of its rotation or mirrored orientation, effectively doubling the structural variety of the training samples.

ii. Photometric and Color Jittering

Random change in brightness, saturation, and hue simulates diverse lighting conditions and camera sensor profiles to prevent the model from overfitting to specific color temperatures. The implementation uses Color Jitter to adjust brightness (0.3), saturation (0.2), and hue (0.1). These adjustments simulate different exposure settings and environmental lighting, forcing the network to rely on shape and form rather than specific pixel intensities.

iii. Gaussian Blur

Gaussian Blur with a variable sigma range between 0.1 and 5.0 is used to simulate out-of-focus or motion blurred images. By softening sharp edges and fine textures, the model is pressured to learn high-level, robust structural features that are essential for distinguishing between 101 separate categories in real world, imperfect conditions.

b. Training Parameter

i. Learning Rate and Warmup Phase

The model is configured with a base learning rate of 7×10^{-4} . To prevent large, destabilizing gradient updates at the start of training a linear warmup ratio of 0.05 is employed. This means that for the first 5% of the total training steps, the learning rate increases linearly from zero to 1×10^{-3} . This gradual introduction of the learning rate allows for stable weight adaptation before the primary training phase begins.

ii. Cosine Learning Rate Scheduling

Following the warmup period, the learning rate is adjusted using a cosine decay schedule over the 12-epoch training duration. Unlike a step-decay approach, the cosine scheduler continuously reduces the learning rate following a half-cosine curve. This results in higher learning rates during the middle of training to facilitate the discovery of optimal loss regions, followed by a rapid but smooth reduction toward the end. This ensures that the model can perform fine-grained weight updates as it approaches a local minimum, leading to a more stable final state.

iii. Label Smoothing Regularization

A label smoothing factor of 0.1 is incorporated into the cross-entropy loss calculation. Standard classification targets use "hard" labels (assigning a probability of 1.0 to the correct class and 0.0 to all others), which can force the model to become overconfident and potentially overfit. Label smoothing redistributes 10% of the target probability mass uniformly across all 101 classes.

This modification changes the optimization objective: the model is no longer encouraged to push the logit of the correct class toward infinity. Instead, it is penalized if its confidence in the correct class exceeds 0.9. This serves as a structural regularizer that prevents the network from memorizing training samples and encourages the learning of more discriminative, generalizable features between the 101 distinct categories.

3. Results



Figure 1 demonstrates a standard convergence pattern over the 10-epoch duration. The training loss exhibits a steep initial decline during the first epoch, dropping from above 4.0 to approximately 1.5, before following a continuous, gradual descent to stabilize near 0.15 by epoch 10. The validation loss tracks a similar downward trajectory, decreasing from approximately 1.8 at epoch 1 to a plateau near 0.5 spanning epochs 8 through 10. A divergence between the training and validation loss curves becomes

visually apparent after epoch 3. While this gap widens as training progresses, the validation loss maintains a relatively flat trajectory in the final epochs rather than exhibiting a sharp upward slope, indicating that the model has not yet entered a state of severe overfitting.

Validation accuracy graph displays consistent incremental gains throughout the majority of the training cycle. The accuracy ascends from a baseline of approximately 0.54 at epoch 1 to a peak of 0.90 at epoch 9. This upward progression includes distinct plateaus, notably between epochs 3 and 4, and epochs 7 and 8, before reaching its maximum observed value. At epoch 10, the validation accuracy records a minor regression, dropping to approximately 0.89.

Analyzing both graphs concurrently indicates that the model reached its maximum capacity for generalization at epoch 9. This is evidenced by the peak validation accuracy aligning with the stabilized minimum of the validation loss. The subsequent decrease in validation accuracy at epoch 10, paired with the stagnant validation loss, suggests the onset of marginal performance degradation on unseen data. Consequently, the weights derived from epoch 9 represent the optimal configuration achieved during this specific training run.

4. Reference

Bello, I., Fedus, W., Du, X., Cubuk, E.D., Srinivas, A., Lin, T.Y., Shlens, J., Zoph, B.: Revisiting ResNets: Improved training and scaling strategies. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 11531-11541 (2021)

Github Link: <https://github.com/Nrjtiii/visdl>